Liposome Nanoparticles

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Abstract

Liposome nanoparticles have become popularized for their use in the biomedical field. Liposomes have many uses including as protien and DNA carriers, but the most common use being drug delivery. Different production methods create varying types and sizes of liposomes. In this paper the extrusion method is used and the number of extrusions is varied from 5-15, using only the odd number of extrusions. The particle diameter and polydispersity index (PDI) is examined based on the number of extrusions the goal being a diameter closest to the filter pore size of 100 nm and the lowest PDI possible. It was found that with increasing number of extrusions the diameter size get closer to 100 nm as well as the PDI value decreases. It was shown that only at 13 and 15 extrusions was the PDI value lower than .2, the industry standard. Another trial at 13 extrusions was done to compare two different brands of filter, Sterlitech and Whatman, which both have the same pore diameter. While the Whatman brand is more expensive meaning high quality control, the results do not prove this, as the Whatman filter had a much larger liposome diameter size. Future experiments should be done at different extrusion values for the Whatman filter to find the optimal extrusion number. As well as more experiments at higher extrusion values to determine if the diameter and PDI trend continues at higher extrusions.

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1 Introduction

Liposome particles are synthetically made capsules of a phospholipid bilayer. These particles are useful in many aspects of biological systems. Firstly, they are able to attach to antibodies, protein receptors and other biosensor molecules. Also, as a capsule they are able to store various contents such as enzymes, proteins, DNA, and drug molecules. There are different methods of production that lead to a highly homogeneous sample of nanoparticles with a small size polydispersity.[1] These particles are able to be produced in varying sizes ranging from tens to thousands of nanometers. This means that size is not a limiting factor on their therapeutic benefit.[2] When these capsules are filled with varying molecules and chemical reactants, cellular processes and chemical reactions can occur before release into the body. When the final product is ready for release, a change of temperature below that of lipid phase transformation or a pulse from a strong electric field will break the membrane apart releasing its contents.[3] These liposome particles have been consistently used for drug delivery especially for cancer treatments. Some drawbacks of their use is the fast elimination of the liposomes from the bloodstream. This happens as with other nanoparticles, they are readily taken up by the MPS system in the liver and spleen, which reduces their circulatory properties. Research has been done to attach ligands to the surface of the liposomes that will attach and bind to cells of interest and are able to accumulate in the target site. These liposomes are used for accumulation of drugs in patients with a specific area of disease, for example cancer patients that have solid tumors.[1] Another variety, long-circulating liposomes are coated on the surface with polymers, such as polyethylene glycol (PEG), to form a protective layer to delay the recognition by opsonins.[1] Further research has focused on making the PEG coating removable to facilitate capture after the drug delivery as well as investigation of other polymers that could be used.

2 Background

Liposomes have unique structures with properties that are ideal for drug delivery. The chemical structure allows higher release efficiency and less cytotoxicity with the ester bond-linked liposomes.[4] They contain phospholipid bilayers which contain a hydrophilic, hydrophobic, and a bridging (linker) domain. The uses of these nanoparticles have increased in recent years, expanding into even the food industry.[5] Its main uses in food are to carry proteins as they can greatly alter the cooking trajectory and outcome, in addition to the overall flavoring and nutritional properties. Similar to drug delivery, the liposome can assist food spoilage and degradation from how the contents are encapsulated. This is due to the plant cell wall structures which can control gel melting, settling, and texture.[5] There is also further research which depends on a low PDI and nanoparticle size for the consistency to build upon. Surface reactions on these nanoparticles can be done to prepare active targeting, which uses ligand-receptor binding.[6] This allows more control on when and where the drugs should be released in drug delivery. The lipid nanoparticles are required for more stable environments and require precise control of the surface density, which can be a challenge when dealing with higher PDI values or larger nanoparticle sizes.

Here, we show the how the PDI and nanoparticle size change from the effects of changing the number of extrusions, from values of 5 up to 15 in increments of 2, in liposome nanoparticle production and using two different brands of filters.

3 Theory

For creating the lipid solution and film, the conservation of mass equation gives us,

$$H_1V_1 = H_2V_2 \tag{1}$$

where H_1 is the initial concentration of DLPC in chloroform solution, V_1 is the initial volume of the solution, and the second set is for the final concentration and volume.

The nanoparticles are created from the solution by pushing through a filter, called an extrusion. Approximately 5-15 extrusions are performed to separate the nanoparticles from the extra material to purify and reduce the size of the liposome nanoparticles.

The liposome nanoparticle is evaluated by its diameter and volume fraction. Nanoparticles are often evaluated by volume fraction,

$$\Phi = \frac{V_i}{V} \tag{2}$$

because it compares the volume of component before mixing, V_i and volume of all components, V, where surface particles become more prominent in a nanoparticle.

The Dynamic Light Scattering (DLS) machine provides a measurement for the nanoparticles by shining rays into the cubicle to measure the absorption at different wavelengths.

This gives us a Polydiversity Index (PDI) that measures the uniformity of our sample after a certain number of extrusions, given by

$$PDI = (\frac{\sigma}{S})^2 \tag{3}$$

where PDI is the polydiversity index of the batch of particles, σ is the standard deviation of an individual peak in size distribution, and **S** is the mean value of an individual peak. This value ranges from 0 to 1, where lower numbers inidicate a more thin and monodisperse solution of particles.

4 Methods

For this experiment a lipid solution was mixed together into a glass vial using phospholipid stock, cholesterol stock and chloroform, 450 μL, 50 μL, and 1.5 mL of each was used respectively. After combining the solution the chloroform was evaporated out facilitated by a stream of air. After only the lipid film remained, it was rehydrated with 2mL of DI water. To aid in the removal of the lipid film from the vial walls the vial was agitated using a vortex mixer for approximately 30 seconds followed by bathing in a sonicator for approximately 2 minutes. After the new solution was thoroughly mixed, around 1 ml was added to one of the syringes of the extruder assembly. For trials 1-6, the extruder was fitted with Sterlitech polycarbonate membrane filter with a pore diameter of .1 micron. For these trials the number of extrusions was increased starting with 5 extrusions and ending with 15, using only an odd number of extrusions. For the final trial 13 extrusions was chosen as it was the trial with the smallest diameter. This trial was fitted with a Whatman Nuclepore Track-Etch Membrane Filter which also has a pore size of 100 nm. This filter is much more expensive than the Sterlitech filter and higher quality control. After extrusion the sample was taken and using a dynamic light scattering (DSL) instrument the particle diameter and PDI were calculated 3 times by the instrument which were then averaged.

5 Results and Discussion

Particle diameter and PDI were measured using the Zetasizer DLS instrument. Statistics were taken from sets of 12 to 33 measurements for each trial. Derived count rates ranged from 672 kcps to 3017 kcps. All measurements were performed at temperatures within 0.1°C of 25°C.

Figure 1 shows that the diameter of liposome particles decreased from 154 nm with five extrusions to 107 nm at 15 extrusions. Since the target size of the liposome particles was 100 nm, the size of liposome particles at 15 extrusions was as expected, but due to the larger diameters at 5 and 9 extrusions, the trend in size was steeper than expected. The size of the particles at 9 extrusions is anomalously high considering the effect of performing further extrusions on particle size diminishing at higher numbers of extrusions. For this trial, a small peak in the size distribution was observed between a diameter of 4000 nm and 6000 nm, indicating impurities in the sample. These impurities can account for the large diameter measured for this trial. Such impurities were likely caused by procedural errors in manual extrusion such as the assembly of the extruder and the sample leaking during extrusion leading to contamination. At 5 extrusions, a broad peak in the number distribution was also observed at the same particle size in addition to a broad area between 1000 and 7000 nm in one of the measurement sets. Due to the low number of extrusions of this trial, the impurities and large average particle size was as expected.

Figure 2 shows the PDI decreasing from 0.258 at 5 extrusions to 0.093 at 15 extrusions with outliers at 0.276 at 9 extrusions and 0.227 at 11 extrusions. The PDI falls under 0.2 beginning with 13 extrusions. The outliers can be attributed to the samples for 9 and 11 extrusions being made separately from those for the other trials, where some error in the procedure can account for the larger PDI. Clogging of the membrane could have occurred due to larger particles being pushed through with too much force applied. Changes in pore sizes in the membrane caused

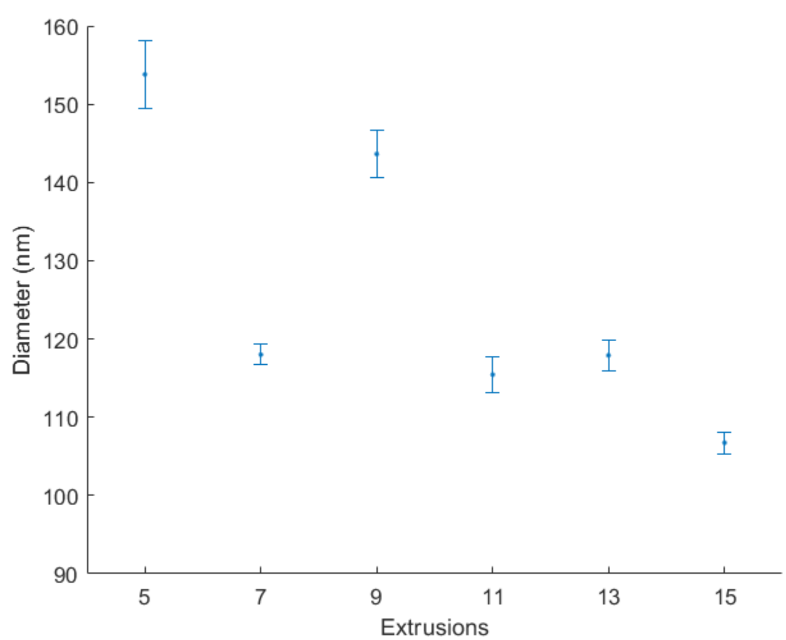

Figure 1: The average diameter of liposome particles with standard deviation was plotted for each number of extrusions. The average diameter generally decreases when increasing the number of extrusions.

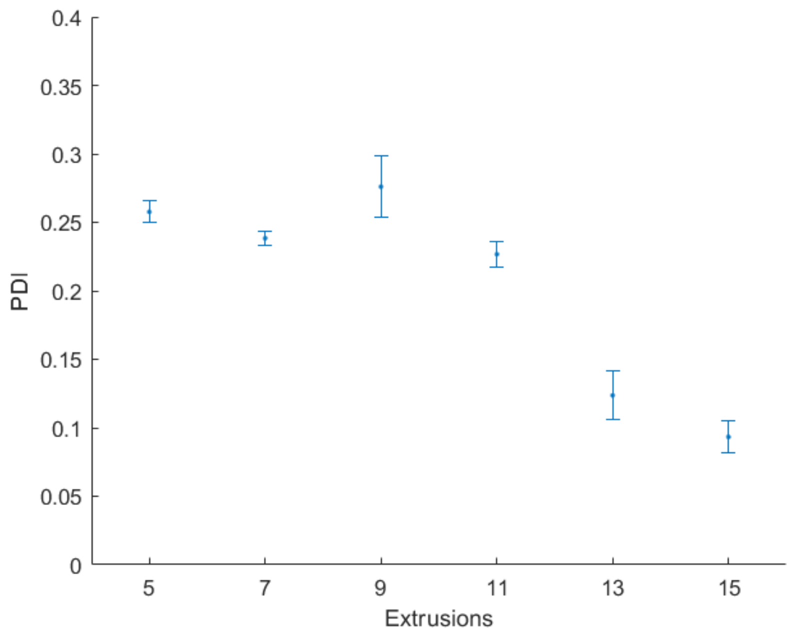

Figure 2: The PDI for each trial was plotted for each number of extrusions. The PDI generally decreases when increasing the number of extrusions

by forcing the sample through it could have occurred, causing a wider distribution of particle sizes to result. Another source of procedural error could have taken place where not enough time was spent with the sample in the vortex mixer and the sonicator. Insufficient agitation of the sample could have led to larger particles being preserved along with the smaller particles present, resulting in a greater PDI. Due to these outliers the PDI trend between 7 extrusions and 13 extrusion is unclear; the samples may or may not reach the desired PDI of below 0.2 at these points.

Figure 3 and Figure 4 show the performance of the Sterlitech brand filter intended for a target nanoparticle size of 100 nm and a Whatman brand filter intended for the same nanoparticle size. Figure 3 shows that for 13 extrusions, the Sterlitech filter yields particles within 20% of the target particle diameter while the Whatman filter yields particles far above the target size. Figure 4 shows that the range of measurements taken was broad for the Whatman filter while the PDI for the Sterlitech filters was on average higher than that of the Whatman filter, though the PDI for both brands fall under the desired range. It is noted that one set of measurements

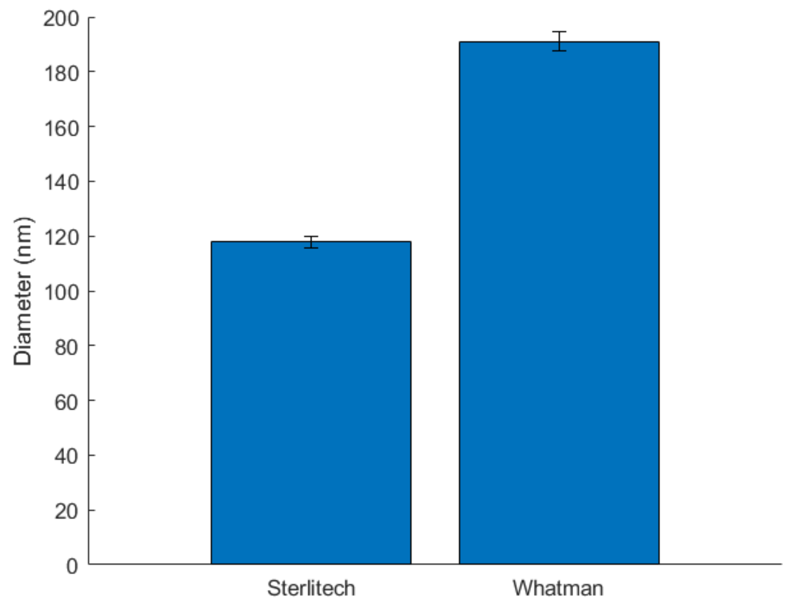

Figure 3: Comparison between average particle sizes yielded by each filter brand. 13 extrusions were done for both trials.

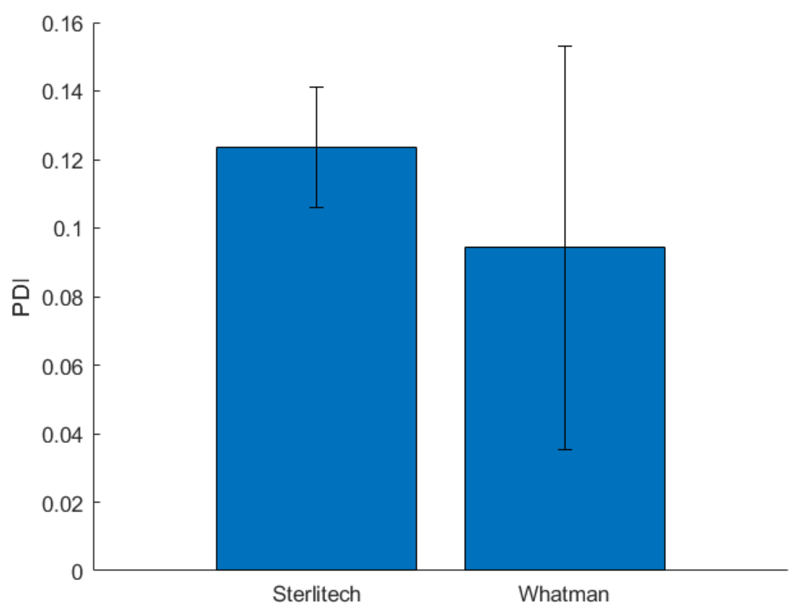

Figure 4: Comparison between the PDI of the samples yielded from each filter brand. Though the averge PDI between trial for the Whatman filter was better, greater consistency was achieved with the Sterlitech filters.

for the Whatman filter resulted in a PDI of 0.027, an outlier that could not excluded accoding to a Grubbs test with an alpha of 0.05. The inconsistency in the Whatman filters could be due to procedural measurement errors when using DLS instrument. Because the Whatman filter was only tested at 13 extrusions, it is unknown how close the diameter 191 nm is to actual pore diameters, which should be in a narrow range considering that the PDI is less than 0.2. We have no figure to show that the same degree of closeness of actual liposome particle sizes to pore sizes of the Whatman filter occur at the same number of extrusions as the Sterlitech filter. Further experimentation is required to assess exactly how many extrusion cycles are optimal for the Whatman filter.

6 Conclusions

In this experiment, for the range of extrusion values that were tested, it was found that 15 extrusions was the optimal amount. At 15 extrusions the liposome diameter and the PDI values were the lowest with values of 106.7 ± 1.38 and $.093 \pm .012$ respectively, shown in Tab. 1. It was shown that only 13 and 15 extrusions gave a PDI value lower than .2, the industry standard. A comparison of two filter brands was also done. While the Whatman brand is of higher quality the results did not give this conclusion. The pore size was at 190 nm when the pore size was given as 100 nm. These results show an inaccurate comparison due to the Whatman trial not being optimal due to a procedural error or an issue with the filter. Future experiment should investigate a larger extrusion range to conclude if more extrusions consistently creates dimater values closer to pore size and smaller PDI values. Also, future experiments should test the Whatman filter at more extrusion values to find the optimal number of extrusions.

Bibliography

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Appendix

1. - Author(s): Johnsen, K. B.; Gubergsson, J. M.
   - Year published: 2017
   - Journal name: JCR
   (a) This journal summarizes the differences between extracellular vesicles (EVs) and liposomes.
   (b) Concludes that many characteristics of EVs are superior to liposomes, but this doesn't account for the range of different types of liposomes. In conclusion the two should both be used the medical field.
2. - Author(s): Torchilin V. P.
   - Year published: 2005
   - Journal name: Nat. Rev. Drug Discovery
   (a) This paper decribes recent advances in the techniques for creating liposomes with different varying modifiers for direction to a certain area of the body and stay in the body for longer periods of time.
   (b) New studies that involve using liposomes as carriers of proteins and peptides
3. - Author(s): Zhang L.; Granick S.
   - Year published: 2006
   - Journal name: Nano Lett.
   (a) Mixing liposomes with charged nanoparticles creates more stability. The particles do not fuse and are instead repelled.
4. - Author(s): Gao Y., Liu X.
   - Year published: 2023

- Journal name: Pharmaceutics

    (a) Liposome drugs have great potential in gene therapy once the limitation of liver targeting is broken.

5. - Author(s): Shukla S., Haldorai, Y., Hwang S. K.
   - Year published: 2017
   - Journal name: Frontiers in Microbiology

    (a) Decribes advances in providing rapid and sensitive detection in portable in-field applications.

6. - Author(s): Menon I., Zaroudi M., Zhang Y.
   - Year published: 2022
   - Journal name: Materials Today Advances

    (a) This journal summarizes the process of creating active targeting drug delivery and its challenges in various prepartion methods.

Table 1: Numerical values for each trial for PDI and Diameter including the Standard Deviation (SD) for both.

Etrustion number	Average PDI	PDI SD	Diameter (nm)	Diameter SD
5	.2577	.008	153.8	4.3
7	.238	.0047	118	1.31
9	.276	.0225	143.6	3.01
11	.19567	.00955	126.15	2.382
13	.1236	.0176	117.9	2.012
15	.0933	.0119	106.733	1.3796
13 (Whatman)	.09433	.05886	190.97	3.5133